ESP32-S3

Hardware Design Guidelines



Release master
Espressif Systems
Nov 05, 2026

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This document provides guidelines for the ESP32-S3 SoC.



Schematic Checklist



PCB Layout Design



Download Guidelines



Resources

Chapter 1

Latest Version of This Document

Check the link to make sure that you use the latest version of this document: https://docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32s3/index.html

1.1 About This Document

1.1.1 Introduction

The hardware design guidelines advise on how to integrate ESP32-S3 into a product. These guidelines will help to achieve optimal performance of your product, ensuring technical accuracy and adherence to Espressif ' s standards. The guidelines are intended for hardware and application engineers.

The document assumes that you possess a certain level of familiarity with the ESP32-S3 SoC. In case you lack prior knowledge, we recommend utilizing this document in conjunction with the ESP32-S3 Series Datasheet.

1.1.2 Latest Version of This Document

Check the link to make sure that you use the latest version of this document: https://docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32s3/index.html

1.2 Product Overview

ESP32-S3 is a system on a chip that integrates the following features:

- Wi-Fi (2.4 GHz band)
- Bluetooth® 5 (LE)
- Dual high-performance Xtensa® 32-bit LX7 CPU cores
- Ultra Low Power coprocessor running either RISC-V or FSM core
- Multiple peripherals
- Built-in security hardware
- USB OTG interface
- USB Serial/JTAG Controller

Powered by 40 nm technology, ESP32-S3 provides a robust, highly-integrated platform, which helps meet the continuous demands for efficient power usage, compact design, security, high performance, and reliability. Typical application scenarios for ESP32-S3 include:

- Smart Home
- Industrial Automation
- Health Care
- Consumer Electronics
- Smart Agriculture
- POS Machines
- Service Robot
- Audio Devices
- Generic Low-power IoT Sensor Hubs
- Generic Low-power IoT Data Loggers
- Cameras for Video Streaming
- USB Devices
- Speech Recognition
- Image Recognition
- Wi-Fi + Bluetooth Networking Card
- Touch and Proximity Sensing

For more information about ESP32-S3, please refer to ESP32-S3 Series Datasheet.

Note: Unless otherwise specified, "ESP32-S3" used in this document refers to the series of chips, instead of a specific chip variant.

1.3 Schematic Checklist

1.3.1 Overview

The integrated circuitry of ESP32-S3 requires only 20 electrical components (resistors, capacitors, and inductors) and a crystal, as well as an SPI flash. The high integration of ESP32-S3 allows for simple peripheral circuit design. This chapter details the schematic design of ESP32-S3.

The following figure shows a reference schematic design of ESP32-S3. It can be used as the basis of your schematic design.

Note that Figure *ESP32-S3 Reference Schematic* shows the connection for 3.3 V, quad, off-package SPI flash/PSRAM.

- In cases where 1.8 V or 3.3 V, octal, in-package or off-package SPI flash/PSRAM is used, GPIO33 ~ GPIO37 are occupied and cannot be used for other functions.
- If an in-package SPI flash/PSRAM is used and VDD_SPI is configured to 1.8 V or 3.3 V via the VDD_SPI_FORCE eFuse, the GPIO45 strapping pin no longer affects the VDD_SPI voltage. In these cases, the presence of R1 is optional. For all other cases, refer to ESP32-S3 Chip Series Datasheet > Section *VDD_SPI Voltage Control* > Table *VDD_SPI Voltage Control* to determine whether R1 should be populated or not.
- The connection for 1.8 V, octal, off-package flash/PSRAM is as shown in Figure *ESP32-S3 Schematic for Off-Package 1.8 V Octal Flash/PSRAM*.
- When only in-package flash/PSRAM is used, there is no need to populate the resistor on the SPI traces or to care the SPI traces.

Any basic ESP32-S3 circuit design may be broken down into the following major building blocks:

- *Power supply*
- *Chip power-up and reset timing*
- *Flash and PSRAM*
- *Clock source*

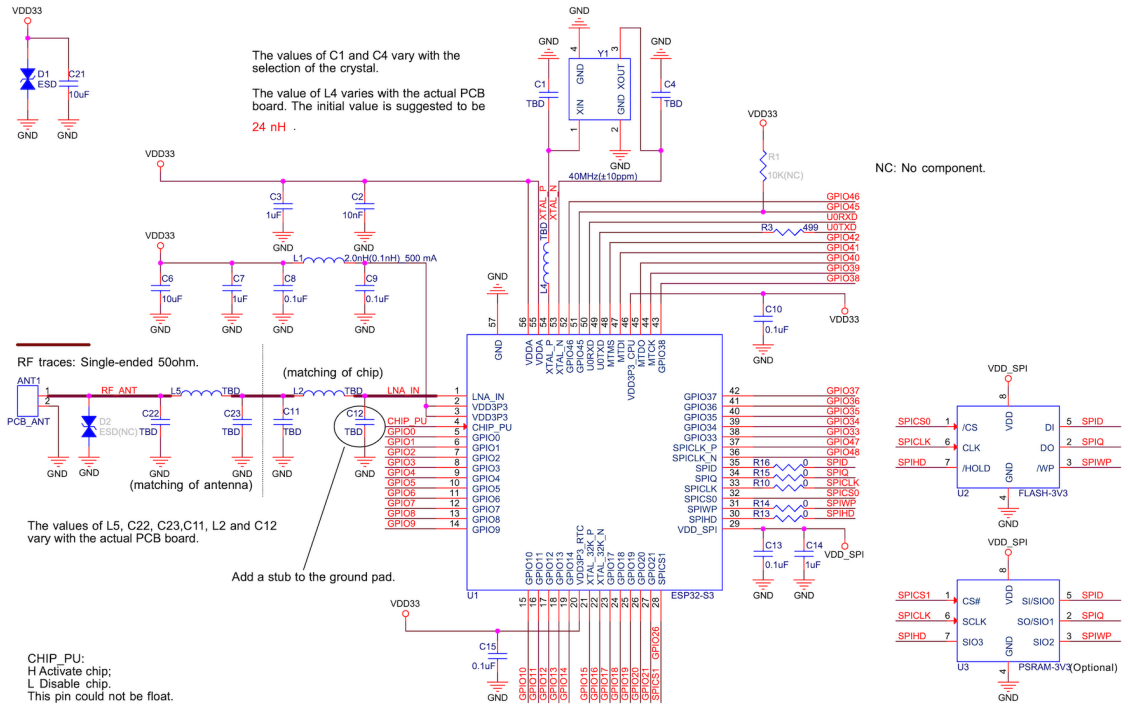


Fig. 1: ESP32-S3 Reference Schematic

- *RF*
- *UART*
- *SPI*
- *Strapping pins*
- *GPIO*
- *ADC*
- *SDIO*
- *USB*
- *Touch sensor*

The rest of this chapter details the specifics of circuit design for each of these sections.

1.3.2 Power Supply

The general recommendations for power supply design are:

- When using a single power supply, the recommended power supply voltage is 3.3 V and the output current is no less than 500 mA.
- It is suggested to add an ESD protection diode and at least 10 µF capacitor at the main power entrance (where the external power supply enters the PCB).

The power scheme is shown in Figure *ESP32-S3 Power Scheme*.

More information about power supply pins can be found in ESP32-S3 Series Datasheet > Section *Power Supply*.

Digital Power Supply

ESP32-S3 has pin46 VDD3P3_CPU as the digital power supply pin, and pin 20 VDD3P3_RTC as the RTC and partial digital power supply pin, with an operating voltage range of 3.0 V ~ 3.6 V. It is recommended to add a 0.1 µF capacitor close to the digital power supply pins in the circuit.

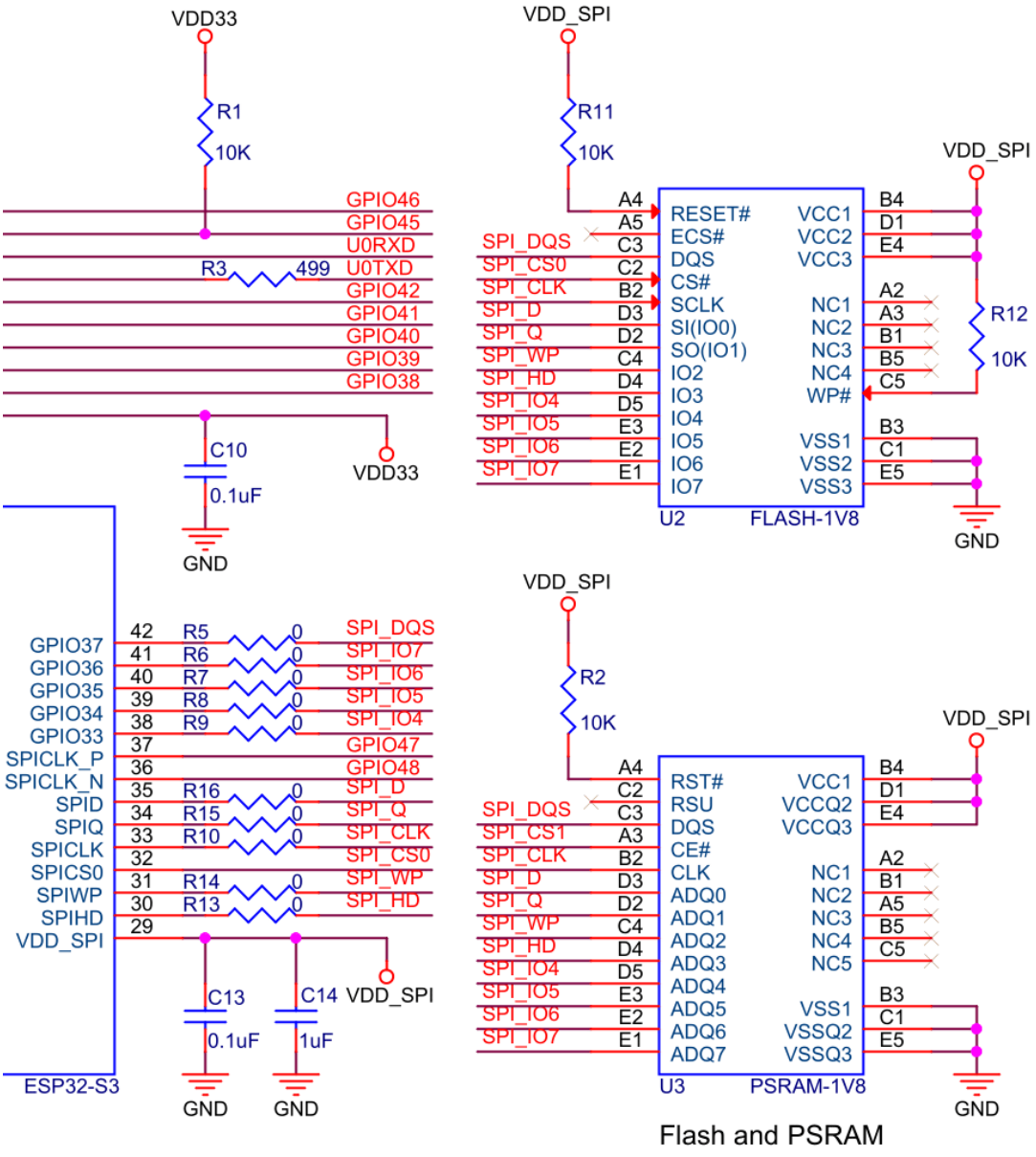


Fig. 2: ESP32-S3 Schematic for Off-Package 1.8 V Octal Flash/PSRAM

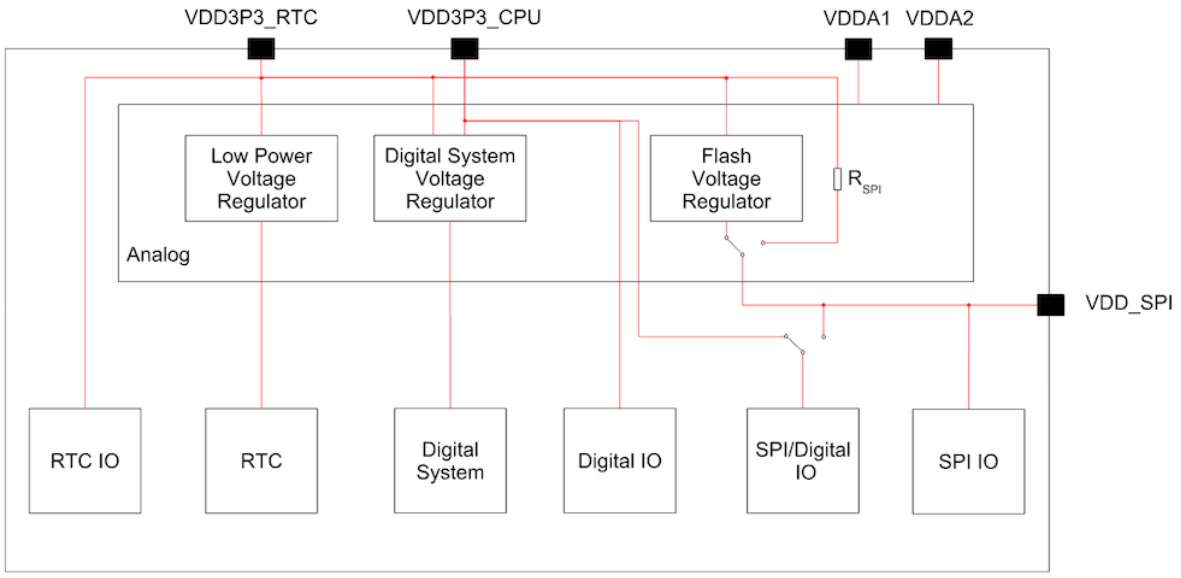


Fig. 3: ESP32-S3 Power Scheme

Pin VDD_SPI serves as the power supply for the external device at either 1.8 V or 3.3 V (default). It is recommended to add extra 0.1 µF and 1 µF decoupling capacitors close to VDD_SPI. Please do not add excessively large capacitors.

- When VDD_SPI operates at 1.8 V, it is powered by ESP32-S3 ' s internal LDO. The typical current this LDO can offer is 40 mA.
- When VDD_SPI operates at 3.3 V, it is driven directly by VDD3P3_RTC through a 14 Ω resistor, therefore, there will be some voltage drop from VDD3P3_RTC.

Attention:

- When using VDDVDD_SPI_SPI as the power supply pin for in-package or off-package 3.3 V flash/PSRAM, please ensure that VDD3P3_RTC remains above 3.0 V to meet the operating voltage requirements of the flash/PSRAM, considering the voltage drop mentioned earlier.
- Note that VDD3P3_RTC cannot supply power alone; all power supplies must be powered on at the same time.

Depending on the value of EFUSE_VDD_SPI_FORCE, the VDD_SPI voltage can be controlled in two ways, as Table *VDD_SPI Voltage Control* shows.

Table 1: VDD_SPI Voltage Control

EFUSE_'	GPIO45	EFUSE_'	Voltage	VDD_SPI Power Source
0	0	Ignored	3.3 V	VDD3P3_RTC via R_{SPI} (default)
0	1	Ignored	1.8 V	Flash Voltage Regulator
1	Ignored	0	1.8 V	Flash Voltage Regulator
1	Ignored	1	3.3 V	VDD3P3_RTC via R_{SPI}

VDD_SPI can also be driven by an external power supply.

It is recommended to use the VDD_SPI output to supply power to external or internal flash/PSRAM.

Analog Power Supply

ESP32-S3 ' s VDD3P3 pins (pin2 and pin3) and VDDA pins (pin55 and pin56) are the analog power supply pins, working at 3.0 V ~ 3.6 V.

For VDD3P3, when ESP32-S3 is transmitting signals, there may be a sudden increase in the current draw, causing power rail collapse. Therefore, it is highly recommended to add a 10 µF capacitor to the power rail, which can work in conjunction with the 1 µF capacitor(s) or other capacitors.

It is suggested to add an extra 10 µF capacitor at the main power entrance. If the main power entrance is close to VDD3P3, then the two 10 µF capacitors can be merged into one.

Add an LC circuit to the VDD3P3 power rail to suppress high-frequency harmonics. The inductor's rated current is preferably 500 mA and above.

For the remaining capacitor circuits, please refer to *ESP32-S3 Reference Schematic*.

1.3.3 Chip Power-up and Reset Timing

ESP32-S3's CHIP_PU pin can enable the chip when it is high and reset the chip when it is low.

When ESP32-S3 uses a 3.3 V system power supply, the power rails need some time to stabilize before CHIP_PU is pulled up and the chip is enabled. Therefore, CHIP_PU needs to be asserted high after the 3.3 V rails have been brought up.

To reset the chip, keep the reset voltage V_{IL_nRST} in the range of (–0.3 ~ 0.25 × VDD3P3_RTC) V. To avoid reboots caused by external interferences, make the CHIP_PU trace as short as possible.

Figure *ESP32-S3 Power-up and Reset Timing* shows the power-up and reset timing of ESP32-S3.

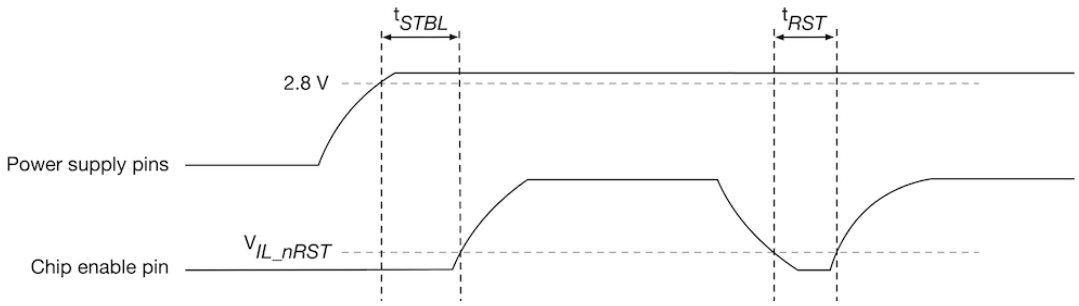


Fig. 4: ESP32-S3 Power-up and Reset Timing

Table *Description of Timing Parameters for Power-up and Reset* provides the specific timing requirements.

Table 2: Description of Timing Parameters for Power-up and Reset

Parameter	Description	Minimum (µs)
t_{STBL}	Time reserved for the power rails to stabilize before the CHIP_PU pin is pulled high to activate the chip	50
t_{RST}	Time reserved for CHIP_PU to stay below V_{IL_nRST} to reset the chip	50

Attention:

- CHIP_PU must not be left floating.
- To ensure the correct power-up and reset timing, it is advised to add an RC delay circuit at the CHIP_PU pin. The recommended setting for the RC delay circuit is usually R = 10 kΩ and C = 1 µF. However, specific parameters should be adjusted based on the characteristics of the actual power supply and the power-up and reset timing of the chip.
- If the user application has one of the following scenarios:
  - Slow power rise or fall, such as during battery charging.
  - Frequent power on/off operations.
  - Unstable power supply, such as in photovoltaic power generation.

> Then, the RC circuit alone may not meet the timing requirements, which may prevent the chip from entering normal operating mode or cause flash erase operations to occasionally fail to complete. In this case, please reserve a power monitor chip to reset the chip when the power supply is abnormal, and the threshold of the power monitor chip is recommended to be around 3.0 V.

1.3.4 Flash and PSRAM

ESP32-S3 requires in-package or off-package flash to store application firmware and data. In-package PSRAM or off-package PSRAM is optional.

In-Package Flash and PSRAM

The tables list the pin-to-pin mapping between the chip and in-package flash/PSRAM. Please note that the following chip pins can connect at most one flash and one PSRAM. That is to say, when there is only flash in the package, the pin occupied by flash can only connect PSRAM and cannot be used for other functions; when there is only PSRAM, the pin occupied by PSRAM can only connect flash; when there are both flash and PSRAM, the pin occupied cannot connect any more flash or PSRAM.

Table 3: Pin-to-Pin Mapping Between Chip and In-Package Quad SPI Flash

ESP32-S3FN8/ESP32-S3FH4R2	In-Package Flash (Quad SPI)
SPICLK	CLK
SPICS0	CS#
SPID	DI
SPIQ	DO
SPIWP	WP#
SPIHD	HOLD#

Table 4: Pin-to-Pin Mapping Between Chip and In-Package Quad SPI PSRAM

ESP32-S3R2/ESP32-S3FH4R2	In-Package PSRAM (2 MB, Quad SPI)
SPICLK	CLK
SPICS1	CE#
SPID	SI/SIO0
SPIQ	SO/SIO1
SPIWP	SIO2
SPIHD	SIO3

Table 5: Pin-to-Pin Mapping Between Chip and In-Package Octal SPI PSRAM

ESP32-S3R8/ESP32-S3R8V	In-Package PSRAM (8 MB, Octal SPI)
SPICLK	CLK
SPICS1	CE#
SPID	DQ0
SPIQ	DQ1
SPIWP	DQ2
SPIHD	DQ3
GPIO33	DQ4
GPIO34	DQ5
GPIO35	DQ6
GPIO36	DQ7
GPIO37	DQS/DM

Off-Package Flash and PSRAM

To reduce the risk of software compatibility issues, it is recommended to use flash and PSRAM models officially validated by Espressif. For detailed model selection, consult the sales or technical support team. If VDD_SPI is used to supply power, make sure to select the appropriate off-package flash and RAM according to the power voltage on VDD_SPI (1.8 V/3.3 V). It is recommended to add zero-ohm resistor footprints in series on the SPI communication lines. These footprints provide flexibility for future adjustments, such as tuning drive strength, mitigating RF interference, correcting signal timing, and reducing noise, if needed.

1.3.5 Clock Source

ESP32-S3 supports two external clock sources:

- *External crystal clock source (Compulsory)*
- *RTC clock source (Optional)*

External Crystal Clock Source (Compulsory)

The ESP32-S3 firmware only supports 40 MHz crystal.

The circuit for the crystal is shown in Figure *ESP32-S3 Schematic for External Crystal*. Note that the accuracy of the selected crystal should be within ±10 ppm.

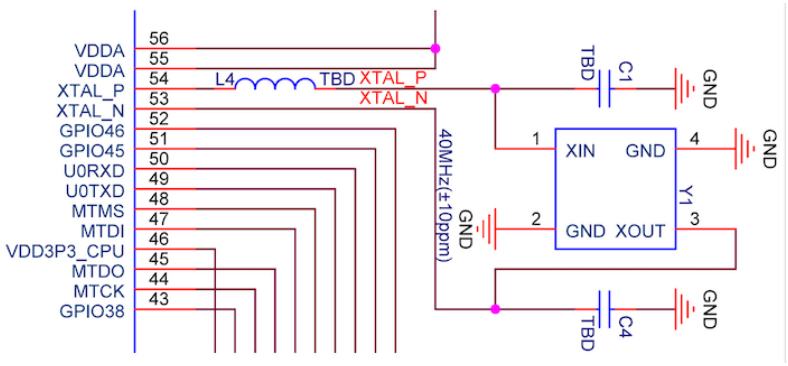


Fig. 5: ESP32-S3 Schematic for External Crystal

Please add a series component on the XTAL_P clock trace. Initially, it is suggested to use an inductor of 24 nH to reduce the impact of high-frequency crystal harmonics on RF performance, and the value should be adjusted after an overall test.

The initial values of external capacitors C1 and C4 can be determined according to the formula:

$$C_L = \frac{C1 \times C4}{C1 + C4} + C_{stray}$$

where the value of C_L (load capacitance) can be found in the crystal's datasheet, and the value of C_{stray} refers to the PCB's stray capacitance. The values of C1 and C4 need to be further adjusted after an overall test as below:

1. Select TX tone mode using the Certification and Test Tool.
2. Observe the 2.4 GHz signal with a radio communication analyzer or a spectrum analyzer and demodulate it to obtain the actual frequency offset.
3. Adjust the frequency offset to be within ±10 ppm (recommended) by adjusting the external load capacitance.

- When the center frequency offset is positive, it means that the equivalent load capacitance is small, and the external load capacitance needs to be increased.
- When the center frequency offset is negative, it means the equivalent load capacitance is large, and the external load capacitance needs to be reduced.
- External load capacitance at the two sides are usually equal, but in special cases, they may have slightly different values.

Note:

- Defects in the manufacturing of crystal (for example, large frequency deviation of more than ±10 ppm, unstable performance within the operating temperature range, etc) may lead to the malfunction of ESP32-S3, resulting in a decrease of the RF performance.
- It is recommended that the amplitude of the crystal is greater than 500 mV.
- When Wi-Fi or Bluetooth connection fails, after ruling out software problems, you may follow the steps mentioned above to ensure that the frequency offset meets the requirements by adjusting capacitors at the two sides of the crystal.

RTC Clock Source (Optional)

ESP32-S3 supports an external 32.768 kHz crystal to act as the RTC clock. The external RTC clock source enhances timing accuracy and consequently decreases average power consumption, without impacting functionality.

Figure *ESP32-S3 Schematic for 32.768 kHz Crystal* shows the schematic for the external 32.768 kHz crystal.

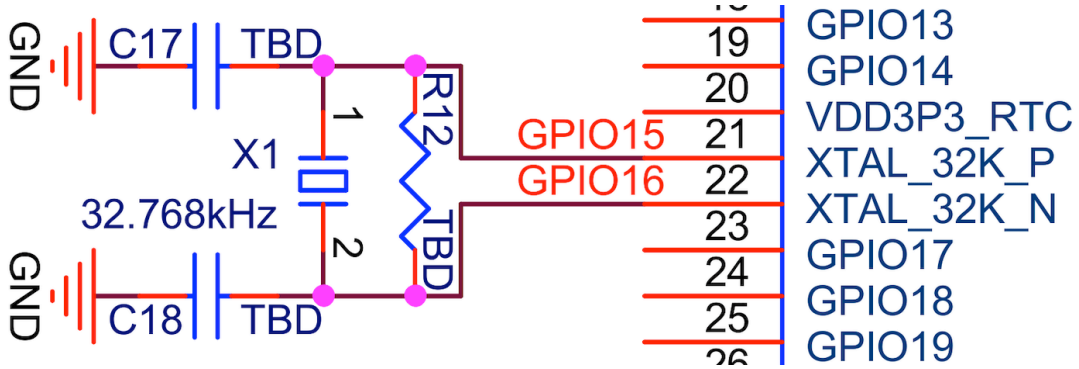


Fig. 6: ESP32-S3 Schematic for 32.768 kHz Crystal

Please note the requirements for the 32.768 kHz crystal:

- Equivalent series resistance (ESR) ≤ 70 kΩ.
- Load capacitance at both ends should be configured according to the crystal's specification.

The parallel resistor R is used for biasing the crystal circuit (5 MΩ < R ≤ 10 MΩ).

In general, you do not need to populate the resistor.

If the RTC clock source is not required, then the pins for the 32.768 kHz crystal can be used as GPIOs.

1.3.6 RF

RF Circuit

ESP32-S3 ' s RF circuit is mainly composed of three parts, the RF traces on the PCB board, the chip matching circuit, the antenna and the antenna matching circuit. Each part should meet the following requirements:

- For the RF traces on the PCB board, 50 Ω impedance control is required.
- For the chip matching circuit, it must be placed close to the chip. A CLC structure is preferred.
  - The CLC structure is mainly used to adjust the impedance point and suppress harmonics.
  - The RF matching circuit is shown in Figure *ESP32-S3 Schematic for RF Matching*.
- For the antenna and the antenna matching circuit, to ensure radiation performance, the antenna ' s characteristic impedance must be around 50 Ω. Adding a CLC matching circuit near the antenna is recommended to adjust the antenna. However, if the available space is limited and the antenna impedance point can be guaranteed to be 50 Ω by simulation, then there is no need to add a matching circuit near the antenna.
- It is recommended to include ESD protection devices for the antenna to mitigate electrostatic interference.

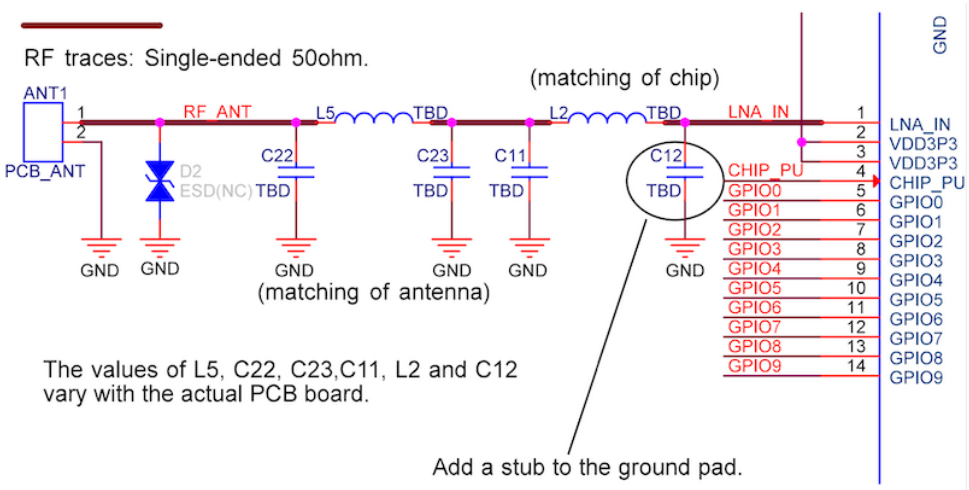


Fig. 7: ESP32-S3 Schematic for RF Matching

RF Tuning

The RF matching parameters vary with the board, so the ones used in Espressif modules could not be applied directly. Follow the instructions below to do RF tuning.

Figure *ESP32-S3 RF Tuning Diagram* shows the general process of RF tuning.

In the matching circuit, define the port near the chip as Port 1 and the port near the antenna as Port 2. S11 describes the ratio of the signal power reflected back from Port 1 to the input signal power, the transmission performance is best if the matching impedance is conjugate to the chip impedance. S21 is used to describe the transmission loss of signal from Port 1 to Port 2. If S11 is close to the chip conjugate point 35+j0 and S21 is less than -35 dB at 4.8 GHz and 7.2 GHz, the matching circuit can satisfy transmission requirements.

Connect the two ends of the matching circuit to the network analyzer, and test its signal reflection parameter S11 and transmission parameter S21. Adjust the values of the components in the circuit until S11 and S21 meet the requirements. If your PCB design of the chip strictly follows the PCB design stated in Chapter *PCB Layout Design*, you can refer to the value ranges in Table *Recommended Value Ranges for Components* to debug the matching circuit.

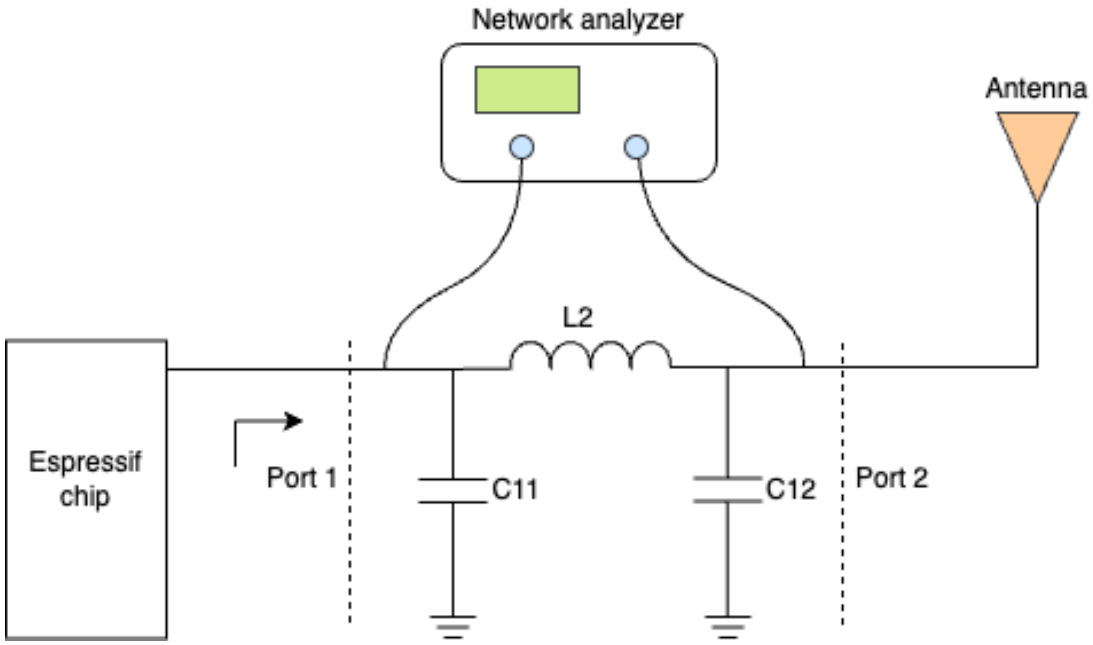


Fig. 8: ESP32-S3 RF Tuning Diagram

Table 6: Recommended Value Ranges for Components

Reference Designator	Recommended Value Range	Serial No.
C11	1.2 ~ 1.8 pF	GRM0335C1H1RXBA01D
L2	2.0 ~ 3.0 nH	LQP03TN2NXB02D
C12	1.8 ~ 1.2 pF	GRM0335C1H1RXBA01D

Please use 0201 packages for RF matching components and add a stub to the first capacitor in the matching circuit at the chip end.

Note: If RF function is not required, it is recommended not to initialize the RF stack in firmware. In this case, the RF pin can be left floating. However, if RF function is enabled, make sure an antenna is connected. Operation without an antenna may result in unstable behavior or potential damage to the RF circuit.

1.3.7 UART

ESP32-S3 includes 3 UART interfaces, UART0, UART1, and UART2. U0TXD and U0RXD are GPIO43 and GPIO44 by default. Other UART signals can be mapped to any available GPIOs by software configurations.

Usually, UART0 is used as the serial port for download and log printing. For instructions on download over UART0, please refer to Section *Download Guidelines*. It is recommended to connect a 499 Ω series resistor to the U0TXD line to suppress harmonics.

For application communication, use UART interfaces other than UART0 if possible. Add a series resistor on the TX line to suppress harmonics.

1.3.8 SPI

When using the SPI function, to improve EMC performance, add a series resistor (or ferrite bead) and a capacitor to ground on the SPI_CLK trace. If space allows, it is recommended to also add a series resistor and capacitor to

ground on other SPI traces. Ensure that the RC/LC components are placed close to the pins of the chip or module.

1.3.9 Strapping Pins

At each startup or reset, a chip requires some initial configuration parameters, such as in which boot mode to load the chip, etc. These parameters are passed over via the strapping pins. After reset, the strapping pins work as normal function pins.

GPIO0, GPIO3, GPIO45, and GPIO46 are strapping pins.

All the information about strapping pins is covered in ESP32-S3 Series Datasheet > Chapter *Boot Configurations*.

For strapping pin information related to VDD_SPI, please refer to Section *Digital Power Supply*.

In this section, we will mainly cover the strapping pins related to boot mode.

After chip reset is released, the combination of GPIO0 and GPIO46 controls the boot mode. See Table *Boot Mode Control*.

Table 7: Boot Mode Control

Boot Mode	GPIO0	GPIO46
Default Config	1 (Pull-up)	0 (Pull-down)
SPI Boot (default)	1	Any value
Joint Download Boot [1]	0	0

[1] Joint Download Boot mode supports the following download methods:

- USB Download Boot:
  - USB-Serial-JTAG Download Boot
  - USB-OTG Download Boot
- UART Download Boot

[2] In addition to SPI Boot and Joint Download Boot modes, ESP32-S3 also supports SPI Download Boot mode. For details, please see ESP32-S3 Technical Reference Manual > Chapter *Chip Boot Control*.

Signals applied to the strapping pins should have specific *setup time* and *hold time*. For more information, see Figure *Setup and Hold Times for Strapping Pins* and Table *Description of Timing Parameters for Strapping Pins*.

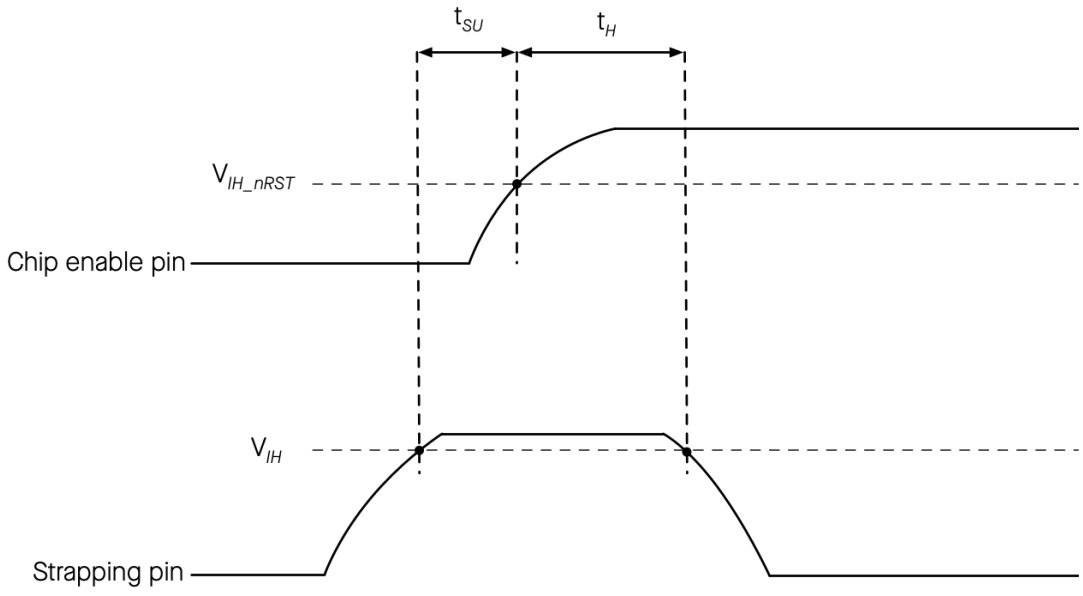


Fig. 9: Setup and Hold Times for Strapping Pins

Table 8: Description of Timing Parameters for Strapping Pins

Parameter	Description	Minimum (ms)
t_{SU}	Time reserved for the power rails to stabilize before the chip enable pin (CHIP_PU) is pulled high to activate the chip.	0
t_{H}	Time reserved for the chip to read the strapping pin values after CHIP_PU is already high and before these pins start operating as regular IO pins.	3

Attention:

- It is recommended to place a pull-up resistor at the GPIO0 pin.
- Do not add high-value capacitors at GPIO0, or the chip may enter download mode.

1.3.10 GPIO

The pins of ESP32-S3 can be configured via IO MUX or GPIO matrix. IO MUX provides the default pin configurations (see ESP32-S3 Series Datasheet > Appendix *ESP32-S3 Consolidated Pin Overview*), whereas the GPIO matrix is used to route signals from peripherals to GPIO pins. For more information about IO MUX and GPIO matrix, please refer to ESP32-S3 Technical Reference Manual > Chapter *IO MUX and GPIO Matrix*.

Some peripheral signals have already been routed to certain GPIO pins, while some can be routed to any available GPIO pins. For details, please refer to ESP32-S3 Series Datasheet > Section *Peripherals*.

When using GPIOs, please:

- Pay attention to the states of strapping pins during power-up.
- Pay attention to the default GPIO configurations after reset (see the table below). For unused pins in the high-impedance state without an internal pull-up or pull-down, it is recommended to add a pull-up or pull-down resistor or enable the internal pull during software initialization to avoid extra power consumption, selecting the direction as required by the external circuit.
- Avoid using the pins already occupied by flash/PSRAM.
- Some pins will have glitches during power-up. Refer to Table *Power-Up Glitches on Pins* for details.
- When USB-OTG Download Boot mode is enabled, some pins will have level output. Refer to Table *IO Pad Status After Chip Initialization in the USB-OTG Download Boot Mode* for details.
- SPICLK_N, SPICLK_P, and GPIO33 ~ GPIO37 work in the same power domain, so if octal 1.8 V flash/PSRAM is used, then SPICLK_P and SPICLK_N also work in the 1.8 V power domain.
- Only GPIOs in the VDD3P3_RTC power domain can be controlled in Deep-sleep mode.

Table 9: IO Pin Default Configuration

No.	Name	Power	At Reset	After Reset
1	LNA_IN			
2	VDD3P3			
3	VDD3P3			
4	CHIP_PU	VDD3P3_RTC		
5	GPIO0	VDD3P3_RTC	IE, WPU	IE, WPU
6	GPIO1	VDD3P3_RTC	IE	IE
7	GPIO2	VDD3P3_RTC	IE	IE
8	GPIO3	VDD3P3_RTC	IE	IE
9	GPIO4	VDD3P3_RTC		
10	GPIO5	VDD3P3_RTC		
11	GPIO6	VDD3P3_RTC		
12	GPIO7	VDD3P3_RTC		

continues on next page

Table 9 – continued from previous page

No.	Name	Power	At Reset	After Reset
13	GPIO8	VDD3P3_RTC		
14	GPIO9	VDD3P3_RTC		IE
15	GPIO10	VDD3P3_RTC		IE
16	GPIO11	VDD3P3_RTC		IE
17	GPIO12	VDD3P3_RTC		IE
18	GPIO13	VDD3P3_RTC		IE
19	GPIO14	VDD3P3_RTC		IE
20	VDD3P3_RTC			
21	XTAL_32K_P	VDD3P3_RTC		
22	XTAL_32K_N	VDD3P3_RTC		
23	GPIO17	VDD3P3_RTC		IE
24	GPIO18	VDD3P3_RTC		IE
25	GPIO19	VDD3P3_RTC		
26	GPIO20	VDD3P3_RTC	USB_PU	USB_PU
27	GPIO21	VDD3P3_RTC		
28	SPICS1	VDD_SPI	IE, WPU	IE, WPU
29	VDD_SPI			
30	SPIHD	VDD_SPI	IE, WPU	IE, WPU
31	SPIWP	VDD_SPI	IE, WPU	IE, WPU
32	SPICS0	VDD_SPI	IE, WPU	IE, WPU
33	SPICLK	VDD_SPI	IE, WPU	IE, WPU
34	SPIQ	VDD_SPI	IE, WPU	IE, WPU
35	SPID	VDD_SPI	IE, WPU	IE, WPU
36	SPICLK_N	VDD_SPI / VDD3P3_CPU	IE	IE
37	SPICLK_P	VDD_SPI / VDD3P3_CPU	IE	IE
38	GPIO33	VDD_SPI / VDD3P3_CPU		IE
39	GPIO34	VDD_SPI / VDD3P3_CPU		IE
40	GPIO35	VDD_SPI / VDD3P3_CPU		IE
41	GPIO36	VDD_SPI / VDD3P3_CPU		IE
42	GPIO37	VDD_SPI / VDD3P3_CPU		IE
43	GPIO38	VDD3P3_CPU		IE
44	MTCK	VDD3P3_CPU		IE
45	MTDO	VDD3P3_CPU		IE
46	VDD3P3_CPU			
47	MTDI	VDD3P3_CPU		IE
48	MTMS	VDD3P3_CPU		IE
49	U0TXD	VDD3P3_CPU	IE, WPU	IE, WPU
50	U0RXD	VDD3P3_CPU	IE, WPU	IE, WPU
51	GPIO45	VDD3P3_CPU	IE, WPD	IE, WPD
52	GPIO46	VDD3P3_CPU	IE, WPD	IE, WPD
53	XTAL_N			
54	XTAL_P			
55	VDDA			
56	VDDA			
57	GND			

- IE –input enabled
- WPU –internal weak pull-up resistor enabled
- WPD –internal weak pull-down resistor enabled
- USB_PU –USB pull-up resistor enabled
  - By default, the USB function is enabled for USB pins (i.e., GPIO19 and GPIO20), and the pin pull-up is decided by the USB pull-up resistor. The USB pull-up resistor is controlled by USB_SERIAL_JTAG_DP/DM_PULLUP and the pull-up value is controlled by USB_SERIAL_JTAG_PULLUP_VALUE. For details, see ESP32-S3 Technical Reference Manual > Chapter *USB Serial/JTAG Controller*.
  - When the USB function is disabled, USB pins are used as regular GPIOs and the pin's internal weak

pull-up and pull-down resistors are disabled by default (configurable by IO_MUX_FUN_WPU/WPD)

Table 10: Power-Up Glitches on Pins

Pin	Glitch[3]	Typical Time (µs)
GPIO1	Low-level glitch	60
GPIO2	Low-level glitch	60
GPIO3	Low-level glitch	60
GPIO4	Low-level glitch	60
GPIO5	Low-level glitch	60
GPIO6	Low-level glitch	60
GPIO7	Low-level glitch	60
GPIO8	Low-level glitch	60
GPIO9	Low-level glitch	60
GPIO10	Low-level glitch	60
GPIO11	Low-level glitch	60
GPIO12	Low-level glitch	60
GPIO13	Low-level glitch	60
GPIO14	Low-level glitch	60
XTAL_32K_P	Low-level glitch	60
XTAL_32K_N	Low-level glitch	60
GPIO17	Low-level glitch	60
GPIO18	Low-level/High-level glitch	60
GPIO19	Low-level glitch/High-level glitch[4]	60
GPIO20	Pull-down glitch/High-level glitch[4]	60

1.3.11 ADC

Table below shows the correspondence between ADC channels and GPIOs.

[3]
- Low-level glitch: the pin is at a low level output status during the time period;
- High-level glitch: the pin is at a high level output status during the time period;
- Pull-down glitch: the pin is at an internal weak pulled-down status during the time period;
- Pull-up glitch: the pin is at an internal weak pulled-up status during the time period.

[4] GPIO19 and GPIO20 pins both have two high-level glitches during chip power-up, each lasting for about 60 µs. The total duration for the glitches and the delay are 3.2 ms and 2 ms respectively for GPIO19 and GPIO20.

Table 11: ADC Functions

GPIO Pin Name	ADC Function
GPIO1	ADC1_CH0
GPIO2	ADC1_CH1
GPIO3	ADC1_CH2
GPIO4	ADC1_CH3
GPIO5	ADC1_CH4
GPIO6	ADC1_CH5
GPIO7	ADC1_CH6
GPIO8	ADC1_CH7
GPIO9	ADC1_CH8
GPIO10	ADC1_CH9
GPIO11	ADC2_CH0
GPIO12	ADC2_CH1
GPIO13	ADC2_CH2
GPIO14	ADC2_CH3
GPIO15	ADC2_CH4
GPIO16	ADC2_CH5
GPIO17	ADC2_CH6
GPIO18	ADC2_CH7
GPIO19	ADC2_CH8
GPIO20	ADC2_CH9

Please add a 0.1 µF filter capacitor between ESP pins and ground when using the ADC function to improve accuracy.

ADC1 is recommended for use.

The calibrated ADC results after hardware calibration and software calibration are shown in the list below. For higher accuracy, you may implement your own calibration methods.

- When ATTEN=0 and the effective measurement range is 0 ~ 850 mV, the total error is ±5 mV.
- When ATTEN=1 and the effective measurement range is 0 ~ 1100 mV, the total error is ±6 mV.
- When ATTEN=2 and the effective measurement range is 0 ~ 1600 mV, the total error is ±10 mV.
- When ATTEN=3 and the effective measurement range is 0 ~ 2900 mV, the total error is ±50 mV.

1.3.12 SDIO

ESP32-S3 only has one SD/MMC Host controller, which cannot be used as a slave device.

The SDIO interface can be configured to any free GPIO by software. Please add pull-up resistors to the SDIO GPIO pins, and it is recommended to reserve a series resistor on each trace.

1.3.13 USB

ESP32-S3 has a full-speed USB On-The-Go (OTG) peripheral with integrated transceivers. The USB peripheral is compliant with the USB 2.0 specification.

ESP32-S3 integrates a USB Serial/JTAG controller that supports USB 2.0 full-speed device.

GPIO19 and GPIO20 can be used as D- and D+ of USB respectively. It is recommended to reserve series resistors (initial value can be 22/33 Ω) and capacitors to ground on the traces (initially can be unpopulated), and place them close to the chip.

The USB RC circuit is shown in Figure *ESP32-S3 USB RC Schematic*.

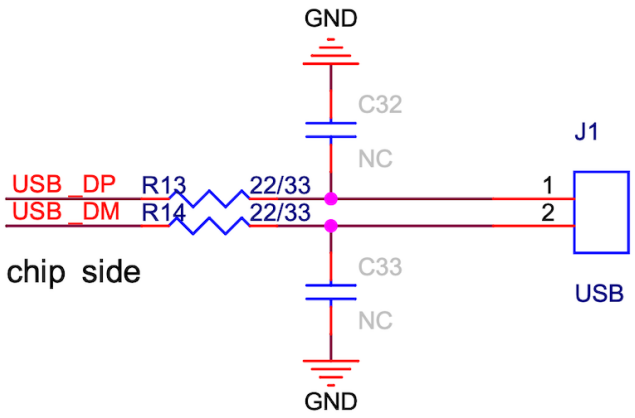


Fig. 10: ESP32-S3 USB RC Schematic

Note that upon power-up, the USB_D+ signal will fluctuate between high and low states. The high-level signal is relatively strong and requires a robust pull-down resistor to drive it low. Therefore, if you need a stable initial state, adding an external pull-up resistor is recommended to ensure a consistent high-level output voltage at startup.

ESP32-S3 also supports download functions and log message printing via USB. For details please refer to Section *Download Guidelines*.

When USB-OTG Download Boot mode is enabled, the chip initializes the IO pad connected to the external PHY in ROM when starts up. The status of each IO pad after initialization is as follows.

Table 12: IO Pad Status After Chip Initialization in the USB-OTG Download Boot Mode

IO Pad	Input/Output Mode	Level Status
VP (MTMS)	INPUT	–
VM (MTDI)	INPUT	–
RCV (GPIO21)	INPUT	–
OEN (MTDO)	OUTPUT	HIGH
VPO (MTCK)	OUTPUT	LOW
VMO(GPIO38)	OUTPUT	LOW

If the USB-OTG Download Boot mode is not needed, it is suggested to disable the USB-OTG Download Boot mode by setting the eFuse bit EFUSE_DIS_USB_OTG_DOWNLOAD_MODE to avoid IO pad state change.

1.3.14 Touch Sensor

ESP32-S3 has 14 capacitive-sensing GPIOs, which detect variations induced by touching or approaching the GPIOs with a finger or other objects. The low-noise nature of the design and the high sensitivity of the circuit allow relatively small pads to be used. Arrays of pads can also be used, so that a larger area or more points can be detected.

The touch sensing performance is further enhanced by the waterproof design and digital filtering feature.

Attention: ESP32-S3 touch sensor has not passed the Conducted Susceptibility (CS) test for now, and thus has limited application scenarios.

Table below shows the correspondence between touch sensor channels and GPIOs.

Table 13: Touch Sensor Functions

GPIO Pin Name	Touch Sensor Function
GPIO1	TOUCH1
GPIO2	TOUCH2
GPIO3	TOUCH3
GPIO4	TOUCH4
GPIO5	TOUCH5
GPIO6	TOUCH6
GPIO7	TOUCH7
GPIO8	TOUCH8
GPIO9	TOUCH9
GPIO10	TOUCH10
GPIO11	TOUCH11
GPIO12	TOUCH12
GPIO13	TOUCH13
GPIO14	TOUCH14

Note that only GPIO14 (TOUCH14) can drive the shield electrode.

When using the touch function, it is recommended to populate a series resistor at the chip side to reduce the coupling noise and interference on the line, and to strengthen the ESD protection. The recommended resistance is from 470 Ω to 2 kΩ, preferably 510 Ω. The specific value depends on the actual test results of the product.

1.4 PCB Layout Design

This chapter introduces the key points of how to design an ESP32-S3 PCB layout using an ESP32-S3 module (see Figure *ESP32-S3 Reference PCB Layout*) as an example.

1.4.1 General Principles of PCB Layout for the Chip

It is recommended to use a four-layer PCB design:

- Layer 1 (TOP): Signal traces and components.
- Layer 2 (GND): No signal traces here to ensure a complete GND plane.
- Layer 3 (POWER): GND plane should be applied to better isolate the RF and crystal. Route power traces and a few signal traces on this layer, provided that there is a complete GND plane under the RF and crystal.
- Layer 4 (BOTTOM): Route a few signal traces here. It is not recommended to place any components on this layer.

A two-layer PCB design can also be used:

- Layer 1 (TOP): Signal traces and components.
- Layer 2 (BOTTOM): Do not place any components on this layer and keep traces to a minimum. Please make sure there is a complete GND plane for the chip, RF, and crystal.

1.4.2 Power Supply

Four-Layer PCB Design

Figure *ESP32-S3 Power Traces in a Four-Layer PCB Design* shows the power traces in a four-layer PCB design.

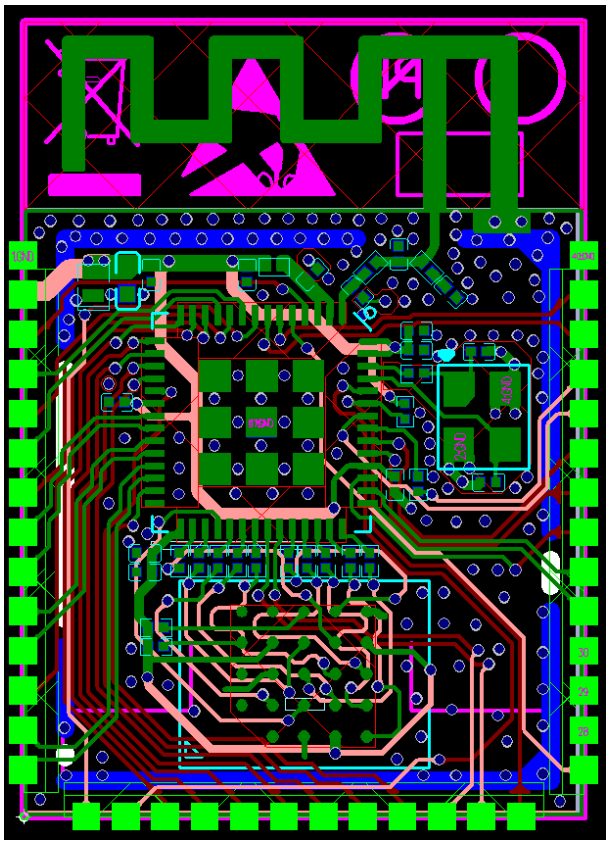


Fig. 11: ESP32-S3 Reference PCB Layout

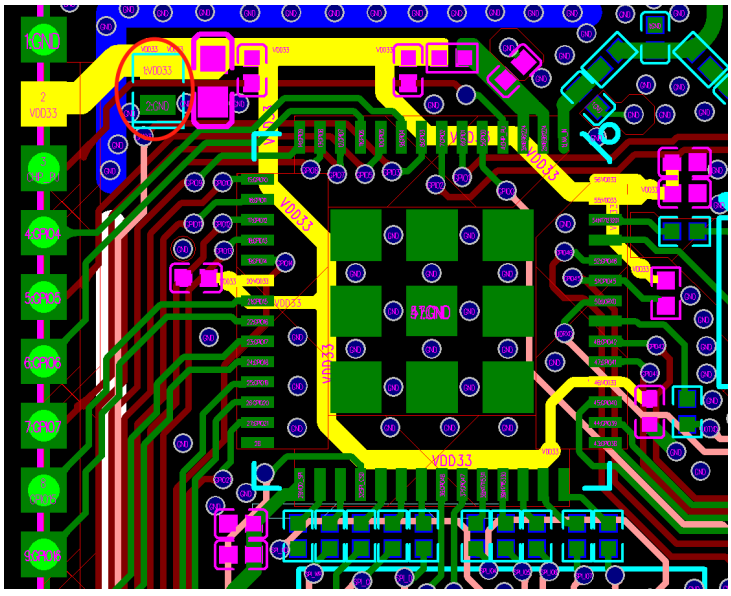


Fig. 12: ESP32-S3 Power Traces in a Four-Layer PCB Design

- A four-layer PCB design is recommended. Whenever possible, route the power traces on the inner layers (not the ground layer) and connect them to the chip pins through vias. There should be at least two vias if the main power traces need to cross layers. The drill diameter on other power traces should be no smaller than the width of the power traces.
- The yellow highlighted traces in Figure *ESP32-S3 Power Traces in a Four-Layer PCB Design* are the 3.3 V power traces. The width of the main power traces should be no less than 25 mil. The width of VDD3P3 at pin2 and pin3 power traces should be no less than 20 mil. The recommended width of other power traces is 10 mil. Ensure the power traces are surrounded by ground copper.
- The red circles in *ESP32-S3 Power Traces in a Four-Layer PCB Design* show ESD protection diodes. Place them close to the power input. Add a 10 µF capacitor before the power trace enters the chip. You can also add a 0.1 µF or 1 µF capacitor in parallel. After that, the power trace can branch out in a star-shaped layout to reduce coupling between different power pins.
- The power supply for pin2 and pin3 is RF related, so please place a 10 µF capacitor for each pin. You can also add a 0.1 µF or 1 µF capacitor in parallel.
- Add a CLC/LC filter circuit near pin2 and pin3 to suppress high-frequency harmonics. The power trace can be routed at a 45-degree angle to maintain distance from adjacent RF traces. Except for the 10 µF capacitor, it is recommended to use 0201 components. This allows the filter circuit for pin2 and pin3 to be placed closer to the pins, with a GND isolation layer separating them from surrounding RF and GPIO traces, while also maximizing the placement of ground vias. Using 0201 components enables placing a via to the bottom layer at the first capacitor near the chip, while maintaining a keep-out area on other layers, further reducing harmonic interference. See Figure *ESP32-S3 Power Traces in a Four-Layer PCB Design*.

- In Figure *ESP32-S3 Power Traces in a Four-Layer PCB Design*, the 10 µF capacitor is shared by the analog power supply VDD3P3 at pin2 and pin3, and the power entrance since the analog power is close to the chip power entrance. If the chip power entrance is not near VDD3P3 at pin2 and pin3, it is recommended to add a 10 µF capacitor to both the chip power entrance and VDD3P3 at pin2 and pin3.
- Place appropriate decoupling capacitors at the rest of the power pins. Ground vias should be added close to the capacitor's ground pad to ensure a short return path.
- The ground pad at the bottom of the chip should be connected to the ground plane through at least nine ground vias.
- The ground pads of the chip and surrounding circuit components should make full contact with the ground copper pour rather than being connected via traces.
- If you need to add a thermal pad EPAD under the chip on the bottom of the module, it is recommended to employ a square grid on the EPAD, cover the gaps with solder paste, and place ground vias in the gaps, as shown in Figure *ESP32-S3 Power Traces in a Four-Layer PCB Design*. This helps effectively reduce solder leakage issues when soldering the module EPAD to the substrate.
- For optimal grounding, connect the EPAD to a large external ground area using wide traces or copper planes. See the figure below.

Two-Layer PCB Design

Figure *ESP32 Power Traces in a Two-Layer PCB Design* shows the power traces in a two-layer PCB design.

- For a two-layer design, ensure to provide a continuous reference ground for the chip, RF, and crystal oscillator, as shown in the figure above.
- In the figure above, the trace VDD33 represents the 3.3 V power trace. Unlike a four-layer design, the power trace should be routed on the top layer as much as possible. Therefore, the thermal pad in the center of the chip should be reduced in size, allowing the power trace to pass between the signal pads and the thermal pad. Vias to the bottom layer should only be used when absolutely necessary.
- Other layout considerations are the same as for a four-layer design.
- Note that there are no official two-layer modules. The figure above uses the ESP32 module as an example.

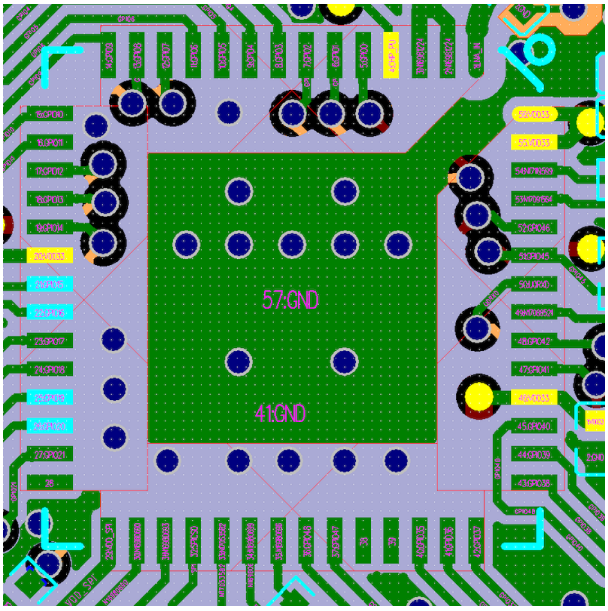


Fig. 13: ESP32-S3 EPAD Design at Chip Bottom

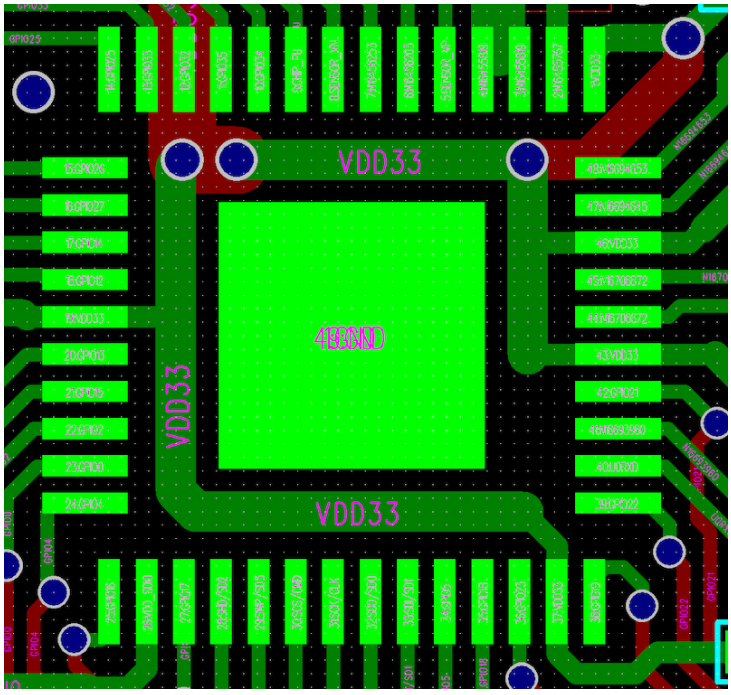


Fig. 14: ESP32 Power Traces in a Two-Layer PCB Design

1.4.3 Crystal

Figure *ESP32-S3 Crystal Layout (with Keep-out Area on Top Layer)* shows a reference PCB layout where the crystal is connected to the ground through vias and a keep-out area is maintained around the crystal on the top layer for ground isolation.

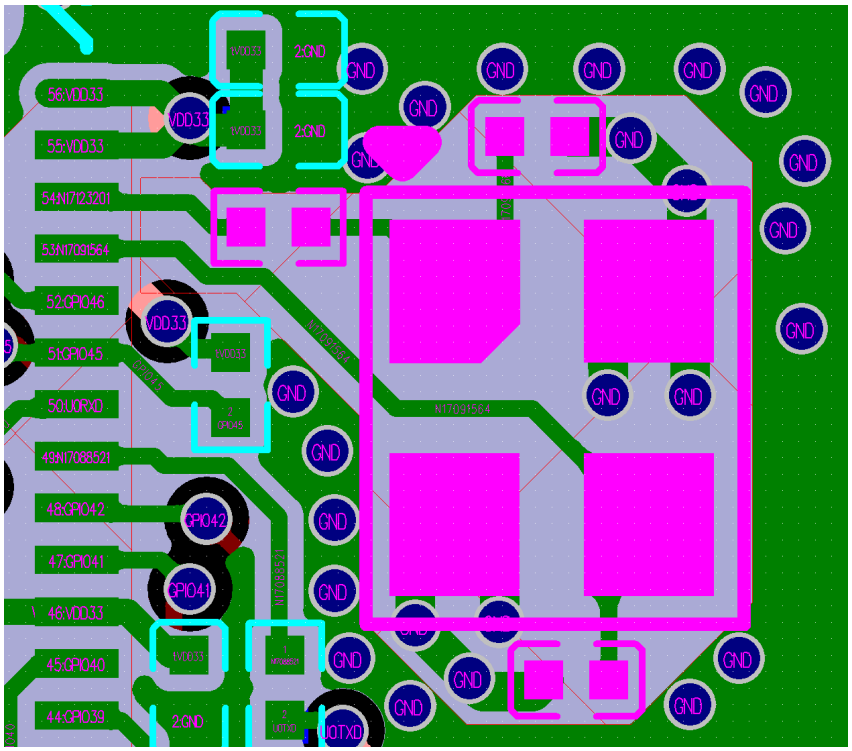


Fig. 15: ESP32-S3 Crystal Layout (with Keep-out Area on Top Layer)

Figure *ESP32-S3 Crystal Layout (without Keep-out Area on Top Layer)* shows the layout for the crystal that is connected to the ground through vias but there is no keep-out area on the top layer for ground isolation.

If there is sufficient ground on the top layer, it is recommended to maintain a keep-out area around the crystal for ground isolation. This helps to reduce the value of parasitic capacitance and suppress temperature conduction, which can otherwise affect the frequency offset. If there is no sufficient ground, do not maintain any keep-out area.

The layout of the crystal should follow the guidelines below:

- Ensure a complete GND plane for the RF, crystal, and chip.
- The crystal should be placed far from the clock pin to avoid interference on the chip. The gap should be at least 2.0 mm. It is good practice to add high-density ground vias stitching around the clock trace for better isolation.
- There should be no vias for the clock input and output traces.
- Components in series to the crystal trace should be placed close to the chip side.
- The external matching capacitors should be placed on the two sides of the crystal, preferably at the end of the clock trace, but not connected directly to the series components. This is to make sure the ground pad of the capacitor is close to that of the crystal.
- Do not route high-frequency digital signal traces under the crystal. It is best not to route any signal trace under the crystal. The vias on the power traces on both sides of the crystal clock trace should be placed as far away from the clock trace as possible, and the two sides of the clock trace should be surrounded by ground copper.
- As the crystal is a sensitive component, do not place any magnetic components nearby that may cause interference, for example large inductance component, and ensure that there is a clean large-area ground plane around the crystal.

1.4.4 RF

The RF trace is routed as shown highlighted in pink in Figure *ESP32-S3 RF Layout in a Four-layer PCB Design*.

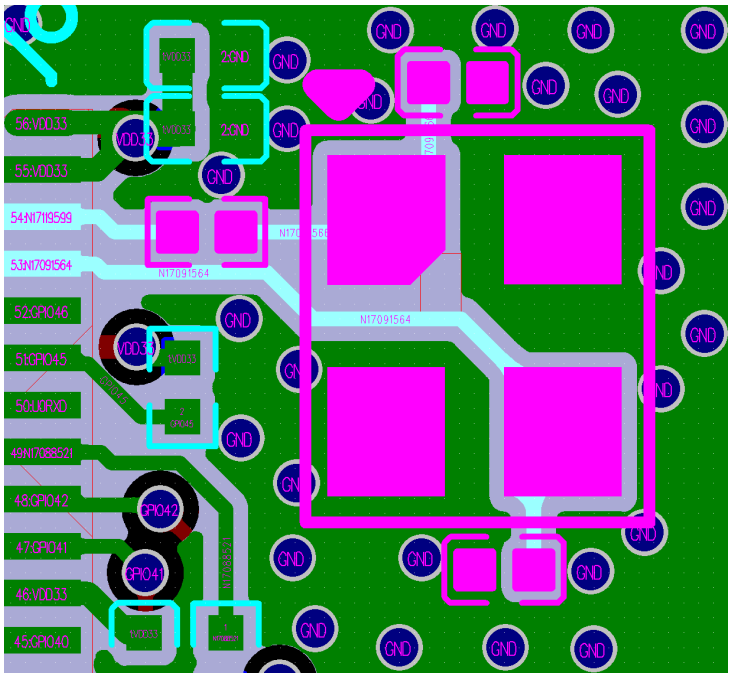


Fig. 16: ESP32-S3 Crystal Layout (without Keep-out Area on Top Layer)

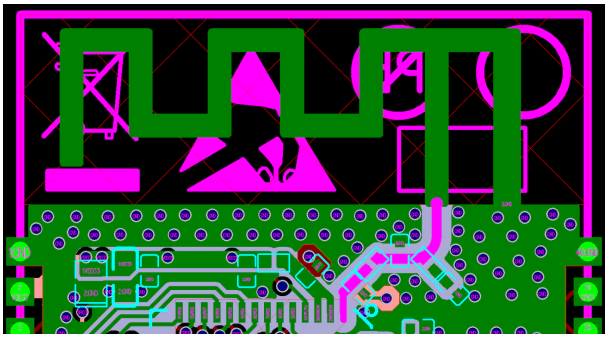


Fig. 17: ESP32-S3 RF Layout in a Four-layer PCB Design

The RF layout should meet the following guidelines:

- The RF trace should have a 50 Ω characteristic impedance. The reference plane is the layer next to the chip. For designing the RF trace at 50 Ω impedance, you could refer to the PCB stack-up design shown below.

Thickness (mm)	Impedance (Ohm)	Gap (mil)	Width (mil)	Gap (mil)
-	50	12.2	12.6	12.2

Stack up	Material	Base copper (oz)	Finished Layer Thickness (mil)	DK
CM			0.4	4
L1_Top	Finished Copper 1 oz	0.33	0.8 (Min)	
PP	7628 TG150 RC50%		8.22	4.6
L2_Gnd		1	1.2	
Core	Core		Adjustable	4.6
L3_Power		1	1.2	
PP	7628 TG150 RC50%		8.22	4.6
L4_Bottom	Finished Copper 1 oz	0.33	0.8 (Min)	
SM			0.4	4

Fig. 18: ESP32-S3 PCB Stack-up Design

- A CLC matching circuit is required for chip tuning. Please use 0201 components and place them close to the pin in a zigzag. In other words, the two capacitors should not be oriented in the same direction to minimize interference.
- Add a stub on the ground pad of the grounding capacitor near the chip side in the matching circuit to suppress the second harmonics. It is preferable to keep the stub length 15 mil, and determine the stub width according to the PCB stack-up so that the characteristic impedance of the stub is 100 Ω ± 10%. The reference plane is the third layer, so the area under the trace on the second layer should be cleared. The trace highlighted in Figure *ESP32-S3 Stub in a Four-layer PCB Design* is the stub. Note that a stub is not required for package types of 0402 and above.
- It is recommended to keep all layers clear under the IPEX antenna connector. See Figure *ESP32-S3 IPEX Antenna Connector Layout*.
- For PCB antennas, make sure to validate them through both simulation and real-world testing on a development board. It is recommended to include an additional CLC matching circuit for antenna tuning. Place this circuit as close to the antenna as possible.
- The RF trace should have a consistent width and not branch out. It should be as short as possible with dense ground vias around for interference shielding.
- The RF trace should be routed on the outer layer without vias, i.e., should not cross layers. The RF trace should be routed at a 135° angle, or with circular arcs if trace bends are required.
- The ground plane on the adjacent layer needs to be complete. Do not route any traces under the RF trace whenever possible.
- There should be no high-frequency signal traces routed close to the RF trace. The RF antenna should be placed away from high-frequency components, such as crystals, DDR SDRAM, high-frequency clocks, etc. In

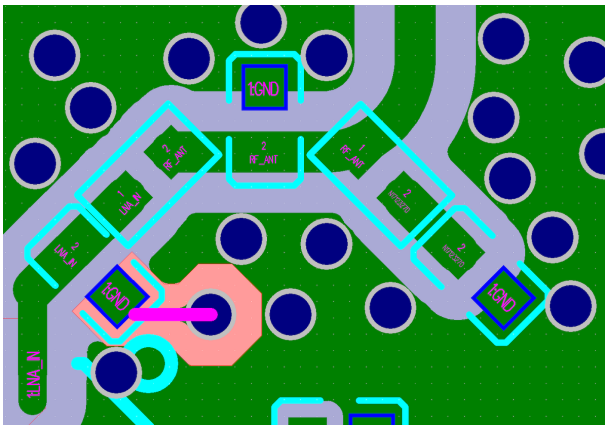


Fig. 19: ESP32-S3 Stub in a Four-layer PCB Design

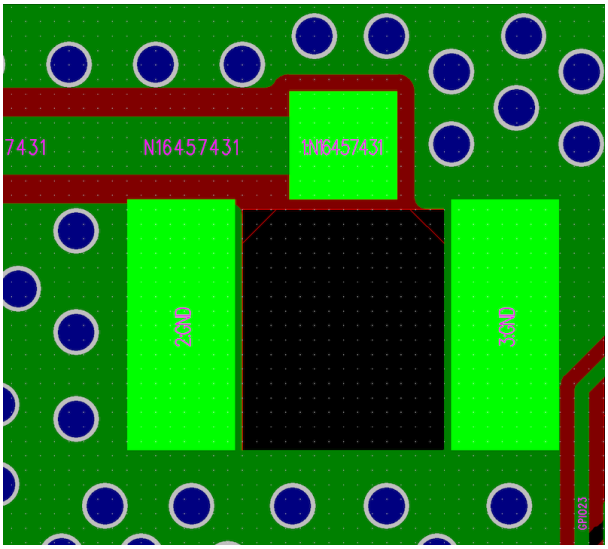


Fig. 20: ESP32-S3 IPEX Antenna Connector Layout

addition, the USB port, USB-to-serial chip, UART signal lines (including traces, vias, test points, header pins, etc.) must be as far away from the antenna as possible. The UART signal line should be surrounded by ground copper and ground vias.

1.4.5 Flash and PSRAM

The layout for flash and PSRAM should follow the guidelines below:

- Place the zero-ohm resistors in series on the SPI lines close to ESP32-S3.
- Route the SPI traces on the inner layer (e.g., the third layer) whenever possible, and add ground copper and ground vias around the clock and data traces of SPI separately.
- Octal SPI traces should have matching lengths.
- If the flash and PSRAM are located far from ESP32-S3, it is recommended to place appropriate decoupling capacitors both at VDD_SPI and near the flash and PSRAM power supply.

Figure *ESP32-S3 Quad SPI Flash Layout* shows the quad SPI flash layout.

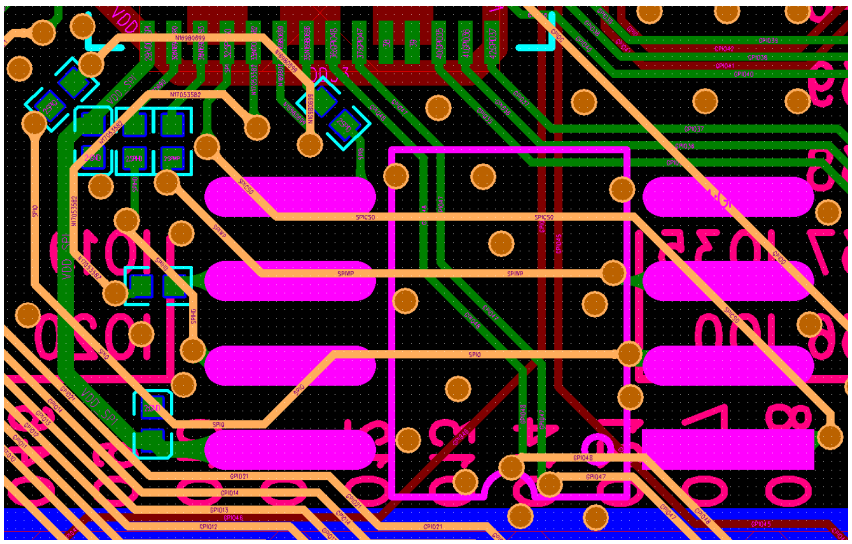


Fig. 21: ESP32-S3 Quad SPI Flash Layout

Figure *ESP32-S3 Octal SPI Flash Layout* shows the octal SPI flash layout.

1.4.6 UART

Figure *ESP32-S3 UART Layout* shows the UART layout.

The UART layout should meet the following guidelines:

- The series resistor on the U0TXD trace needs to be placed close to the chip side and away from the crystal.
- The U0TXD and U0RXD traces on the top layer should be as short as possible.
- The UART trace should be surrounded by ground copper and ground vias stitching.

1.4.7 General Principles of PCB Layout for Modules (Positioning a Module on a Base Board)

If module-on-board design is adopted, attention should be paid while positioning the module on the base board. The interference of the baseboard on the module' s antenna performance should be minimized.

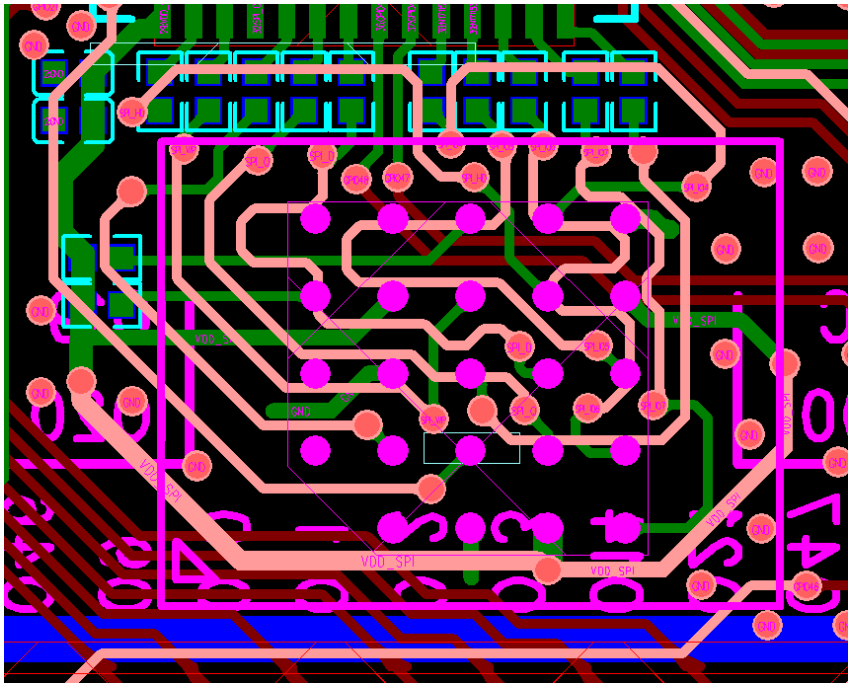


Fig. 22: ESP32-S3 Octal SPI Flash Layout

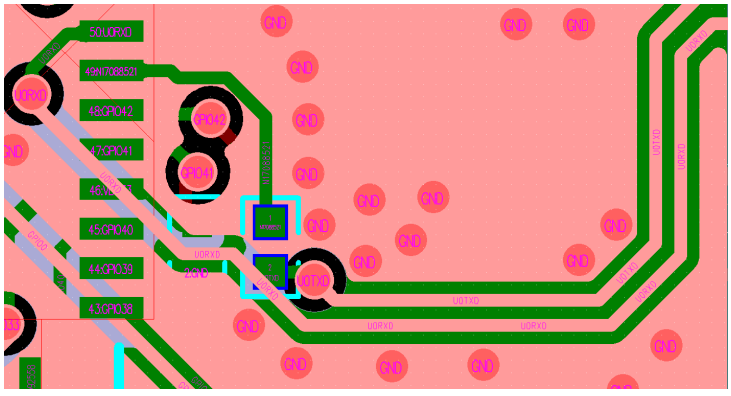


Fig. 23: ESP32-S3 UART Layout

It is suggested to place the module's on-board PCB antenna outside the base board, and the feed point of the antenna close to the edge of the base board. In the following example figures, positions with mark ✓ are strongly recommended, while positions without a mark are not recommended.

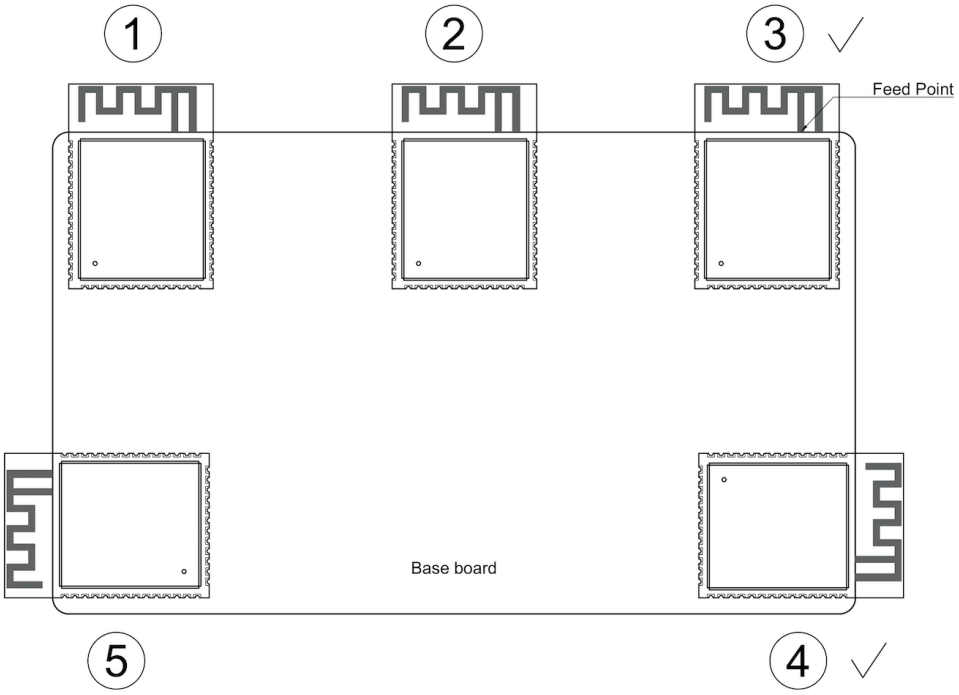


Fig. 24: Placement of ESP32-S3 Modules on Base Board (antenna feed point on the right)

If the antenna cannot extend beyond the board edge, the feed point should still be placed as close to the board edge as possible. Then cut off the base board on both sides of the antenna and below it to minimize the impact of the base board material on the PCB antenna and provide a sufficiently large clearance area for the PCB antenna. Note that the module should not be placed in the center of the board with clearance created by hollowing out on all four sides. Figure *Keepout Zone for ESP32-S3 Module's Antenna (Antenna feed point on the right)* shows the suggested clearance area. Please note that sufficient ground copper and dense ground vias should be placed on the base board near the antenna.

After the base board is placed in the end product, please consider the impact of the housing on the antenna during end-product design. Ensure that the PCB antenna on the base board also has a sufficiently large clearance area inside the housing. A clearance of at least 15 mm is recommended in all directions.

Please note that the final end product should be tested for throughput and communication range to ensure RF performance.

1.4.8 USB

The USB layout should meet the following guidelines:

- Reserve space for resistors and capacitors on the USB traces close to the chip side.
- Use differential pairs and route them in parallel at equal lengths. Maintain a differential pair impedance of 90 Ω with a tolerance of ±10%.
- USB differential traces should minimize via transitions as much as possible to ensure better impedance control and avoid signal reflections. If vias are necessary, add a pair of ground return vias at each transition point.
- Ensure there is a continuous reference layer (a ground layer is recommended) beneath the USB traces.
- Surround the USB traces with ground copper.

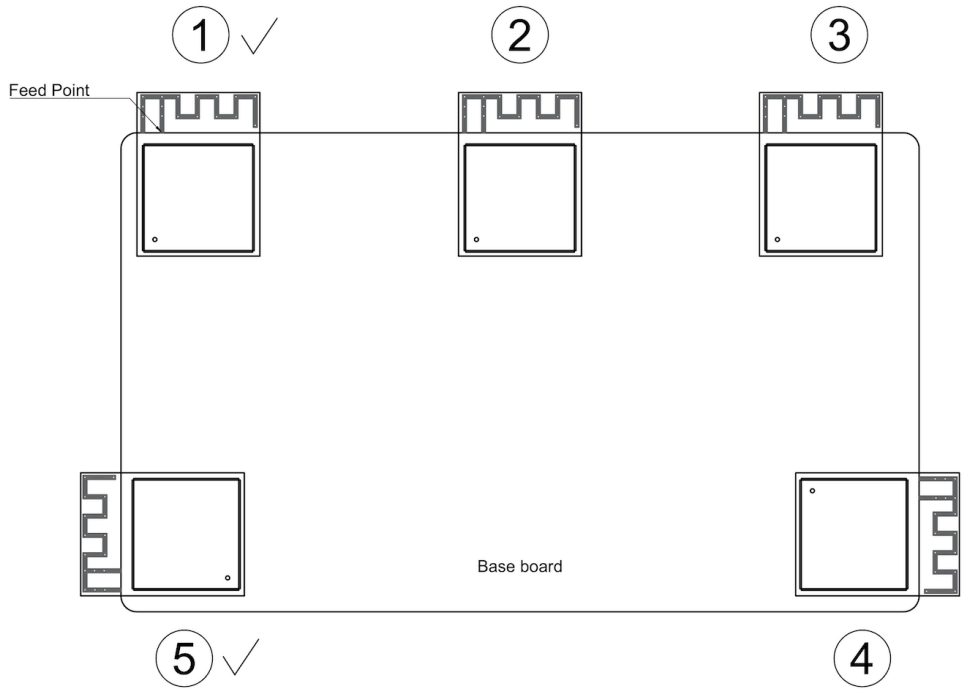


Fig. 25: Placement of ESP32-S3 Modules on Base Board (antenna feed point on the left)

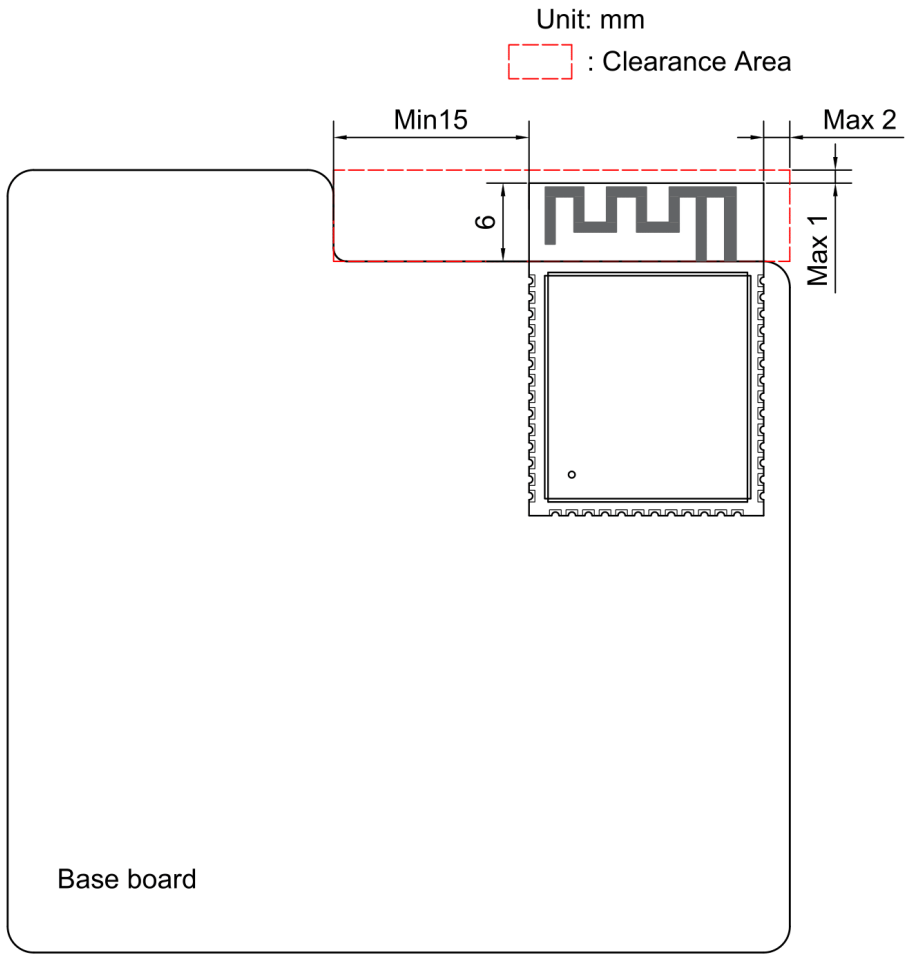


Fig. 26: Keepout Zone for ESP32-S3 Module’ s Antenna (Antenna feed point on the right)

1.4.9 SDIO

The SDIO layout should follow the guidelines below:

- Minimize parasitic capacitance of SDIO traces as they involve high-speed signals.
- The trace lengths for SDIO_CMD and SDIO_DATA0 ~ SDIO_DATA3 should be within ± 50 mil of the SDIO_CLK trace length. Use serpentine routing if necessary.
- For SDIO routing, maintain a 50 Ω single-ended impedance with a tolerance of ±10%.
- Keep the total trace length from SDIO GPIOs to the master SDIO interface as short as possible, ideally within 2000 mil.
- Ensure that SDIO traces do not cross layers. Besides, a reference plane (preferably a ground plane) must be placed beneath the traces, and continuity of the reference plane must be ensured.
- It is recommended to surround the SDIO_CLK trace with ground copper.

1.4.10 Touch Sensor

ESP32-S3 offers up to 14 capacitive IOs that detect changes in capacitance on touch sensors due to finger contact or proximity. The chip's internal capacitance detection circuit features low noise and high sensitivity. It allows to use touch pads with smaller area to implement the touch detection function. You can also use the touch panel array to detect a larger area or more test points.

Figure *ESP32-S3 Typical Touch Sensor Application* depicts a typical touch sensor application.

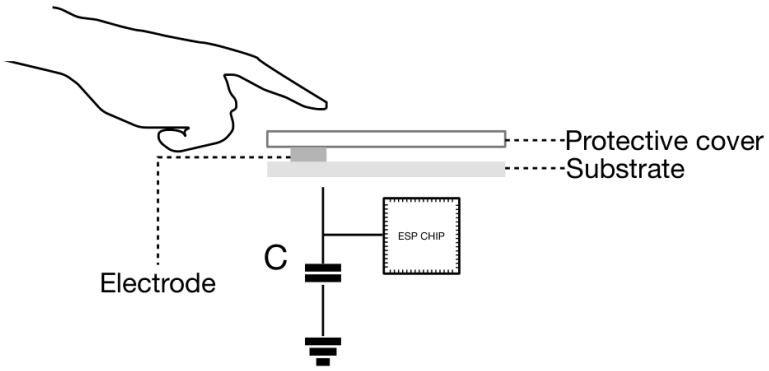


Fig. 27: ESP32-S3 Typical Touch Sensor Application

To prevent capacitive coupling and other electrical interference to the sensitivity of the touch sensor system, the following factors should be taken into account.

Electrode Pattern

The proper size and shape of an electrode improves system sensitivity. Round, oval, or shapes similar to a human fingertip are commonly applied. Large size or irregular shape might lead to incorrect responses from nearby electrodes.

Figure *ESP32-S3 Electrode Pattern Requirements* shows the proper and improper size or shape of electrode. Please note that the examples illustrated in the figure are not of actual scale. It is suggested to use a human fingertip as reference.

PCB Layout

Figure *ESP32-S3 Sensor Track Routing Requirements* illustrates the general guidelines to routing traces. Specifically,

- The trace should be as short as possible and no longer than 300 mm.

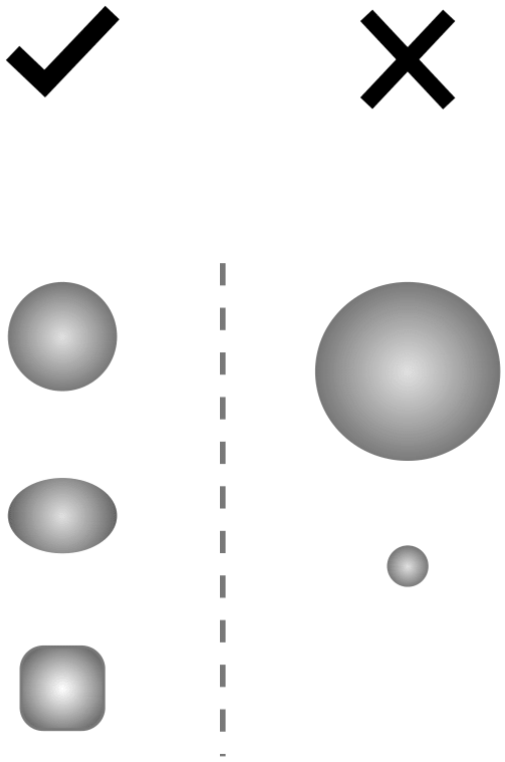


Fig. 28: ESP32-S3 Electrode Pattern Requirements

- The trace width (W) can not be larger than 0.18 mm (7 mil).
- The alignment angle (R) should not be less than 90°.
- The trace-to-ground gap (S) should be in the range of 0.5 mm to 1 mm.
- The electrode diameter (D) should be in the range of 8 mm to 15 mm.
- Hatched ground should be added around the electrodes and traces.
- The traces should be isolated well and routed away from that of the antenna.

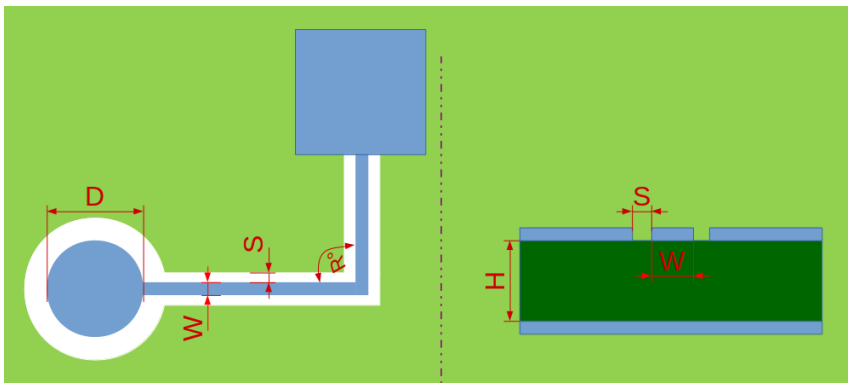


Fig. 29: ESP32-S3 Sensor Track Routing Requirements

Note: For more details on the hardware design of the touch sensor, please refer to Touch Sensor Application Note.

Waterproof and Proximity Sensing Design

ESP32-S3 touch sensor has a waterproof design and features proximity sensor function. Figure *ESP32-S3 Waterproof and Proximity Sensing Design* shows an example layout of a waterproof and proximity sensing design.

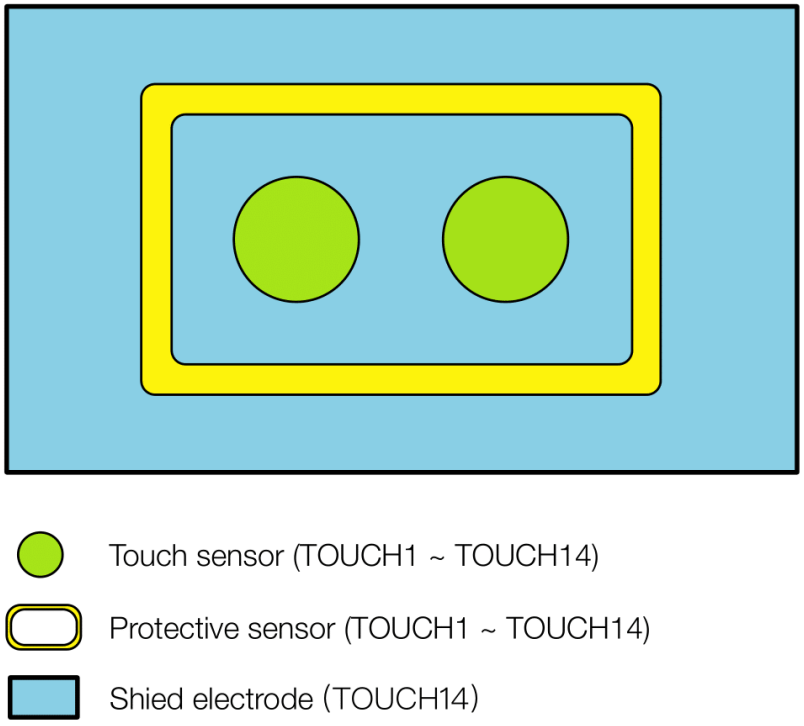


Fig. 30: ESP32-S3 Waterproof and Proximity Sensing Design

Note the following guidelines to better implement the waterproof and proximity sensing design:

- The recommended width of the shield electrode width is 2 cm.
- Employ a grid on the top layer with a trace width of 7 mil and a grid width of 45 mil (25% fill). The filled grid is connected to the driver shield signal.
- Employ a grid on the bottom layer with a trace width of 7 mil and a grid width of 70 mil (17% fill). The filled grid is connected to the driver shield signal.
- The protective sensor should be in a rectangle shape with curved edges and surround all other sensors.
- The recommended width of the protective sensor is 2 mm.
- The recommended gap between the protective sensor and shield sensor is 1 mm.
- The sensing distance of the proximity sensor is directly proportional to the area of the proximity sensor. However, increasing the sensing area will introduce more noise. Actual testing is needed for optimized performance.
- It is recommended that the shape of the proximity sensor is a closed loop. The recommended width is 1.5 mm.

1.4.11 Typical Layout Problems and Solutions

When ESP32-S3 sends data packages, the voltage ripple is small, but RF TX performance is poor.

Analysis: The RF TX performance can be affected not only by voltage ripples, but also by the crystal itself. Poor quality and big frequency offsets of the crystal decrease the RF TX performance. The crystal clock may be corrupted by other interfering signals, such as high-speed output or input signals. In addition, high-frequency signal traces, such as the SDIO traces and UART traces under the crystal, could also result in the malfunction of the crystal. Besides, sensitive components or radiating components, such as inductors and antennas, may also decrease the RF performance.

Solution: This problem is caused by improper layout for the crystal and can be solved by re-layout. Please refer to Section *Crystal* for details.

When ESP32-S3 sends data packages, the power value is much higher or lower than the target power value, and the EVM is relatively poor.

Analysis: The disparity between the tested value and the target value may be due to signal reflection caused by the impedance mismatch on the transmission line connecting the RF pin and the antenna. Besides, the impedance mismatch will affect the working state of the internal PA, making the PA prematurely access the saturated region in an abnormal way. The EVM becomes poor as the signal distortion happens.

Solution: Match the antenna' s impedance with the π-type circuit on the RF trace, so that the impedance of the antenna as seen from the RF pin matches closely with that of the chip. This reduces reflections to the minimum.

TX performance is not bad, but the RX sensitivity is low.

Analysis: Good TX performance indicates proper RF impedance matching. Poor RX sensitivity may result from external coupling to the antenna. For instance, the crystal signal harmonics could couple to the antenna. If the TX and RX traces of UART cross over with RF trace, they will affect the RX performance, as well. If there are many high-frequency interference sources on the board, signal integrity should be considered.

Solution: Keep the antenna away from crystals. Do not route high-frequency signal traces close to the RF trace. Please refer to Section *RF* for details.

1.5 Download Guidelines

You can download firmware to ESP32-S3 via UART and USB.

To download via UART:

1. Before the download, make sure to set the chip or module to Joint Download Boot mode, according to Table *Boot Mode Control*.
2. Power up the chip or module and check the log via the UART0 serial port. If the log shows "waiting for download", the chip or module has entered Joint Download Boot mode.
3. Use the `Flash Download Tool` to flash firmware into flash via UART.
4. After the firmware has been downloaded, pull GPIO0 high or leave it floating to make sure that the chip or module enters SPI Boot mode.
5. Power up the chip or module again. The chip will read and execute the new firmware during initialization.

To download via USB:

1. If the flash is empty, set the chip or module to Joint Download Boot mode, according to Table *Boot Mode Control*.
2. Power up the chip or module and check the log via USB serial port. If the log shows "waiting for download", the chip or module has entered Joint Download Boot mode.
3. Use the `Flash Download Tool` to flash firmware into flash via USB.
4. After the firmware has been downloaded, pull GPIO0 high or leave it floating to make sure that the chip or module enters SPI Boot mode.
5. Power up the chip or module again. The chip will read and execute the new firmware during initialization.
6. If the flash is not empty, start directly from Step 3.

Note:

- For firmware download instructions, see also ESP Product Firmware Download Instructions.
- For how to check serial port output, see also Establish Serial Connection with ESP32-S3.
- It is advised to download the firmware only after the "waiting for download" log shows via the serial port.
- Serial tools cannot be used simultaneously with the Flash Download Tool on one COM port.

- The USB auto-download will be disabled if the following conditions occur in the application, where it will be necessary to set the chip or module to Joint Download Boot mode first by configuring the strapping pin.
  - USB PHY is disabled by the application;
  - USB is configured for other USB functions, e.g., USB host, USB standard device;
  - USB IOs are configured to other peripherals, such as UART and LEDC.
- It is recommended that the user retains control of the strapping pins to avoid the USB download function not being available in case of the above scenario.
- It is recommended to retain the UART download interface, as the current RF test firmware only supports the UART interface.

1.6 Related Documentation and Resources

1.6.1 ESP32-S3 Modules

This section lists only sample and mass-production ESP32-S3 modules. For the complete list, see the Modules page on the Espressif website.

- ESP32-S3-MINI-1
- ESP32-S3-MINI-1U
- ESP32-S3-WROOM-1
- ESP32-S3-WROOM-1U
- ESP32-S3-WROOM-2

For module reference designs please refer to:

- Download links

Note: Use the following tools to open the files in module reference designs:

- .DSN files: OrCAD Capture V16.6
- .pcb files: Pads Layout VX.2. If you cannot open the .pcb files, please try importing the .asc files into your software to view the PCB layout.

For the terms applicable to the use of these reference designs, please refer to *Terms of Use for Reference Designs* in *Disclaimer and Copyright Notice*.

1.6.2 ESP32-S3 Development Boards

This section lists only sample and mass-production ESP32-S3 development boards. For the complete list, see the Development Boards page on the Espressif website.

- ESP-VoCat
- ESP32-S3-BOX-3
- ESP32-S3-DevKitC-1
- ESP32-S3-Korvo-2
- ESP32-S3-LCD-EV-Board
- ESP32-S3-USB-OTG

1.6.3 Other Related Documentation and Resources

- Chip Datasheet (PDF)
- Technical Reference Manual (PDF)
- Chip Errata
- ESP32-S3 Chip Variants
- Espressif KiCad Library
- ESP Product Selector
- Regulatory Certificates
- User Forum (Hardware)
- Technical Support
- ESP-FAQ

1.7 Glossary

The glossary contains terms and acronyms that are used in this document.

Term	Description
CLC	Capacitor-Inductor-Capacitor
DDR SDRAM	Double Data Rate Synchronous Dynamic Random-Access Memory
ESD	Electrostatic Discharge
LC	Inductor-Capacitor
PA	Power Amplifier
RC	Resistor-Capacitor
RTC	Real-Time Clock
Zero-ohm resistor	A zero-ohm resistor acts as a placeholder in the circuit, allowing for the replacement with a higher-ohm resistor based on specific design requirements.

1.8 Revision History

Table 14: Revision History

Date	Version	Release Notes
2025-07-03	v1.8	•***Schematic Checklist*** – Section *Overview*: Updated Figure *ESP32-S3 Reference Schematic* – Section *Power Supply*: Added Figure *ESP32-S3 Power Scheme*; deleted Figure *ESP32-S3 Schematic for Digital Power Supply Pins* and the *RTC Power Supply* section; updated descriptions – Section *External Crystal Clock Source (Compulsory)*: Updated Figure *ESP32-S3 Schematic for External Crystal* – Section *RF Tuning*: Updated descriptions – Section *UART*: Updated descriptions – Section *Strapping Pins*: Updated descriptions – Section *GPIO*: Simplified Table *IO Pin Default Configuration* – Section *ADC*: Added Table *ADC Functions* – Section *USB*: Added Figure *ESP32-S3 USB RC Schematic* – Section *Touch Sensor*: Added Table *Touch Sensor Functions*; updated descriptions •***PCB Layout Design*** – Section *Power Supply*: Restructured the section and updated descriptions – Section *RF*: Updated Figure *ESP32-S3 PCB Stack-up Design*; added Figure *ESP32-S3 IPEX Antenna Connector Layout*; updated descriptions – Section *RF Circuit*: Updated Figure *ESP32-S3 Schematic for RF Matching*
2025-06-05	v1.7	•***PCB Layout Design*** – Section *USB*: Updated descriptions about the USB layout guidelines
2025-05-23	v1.6	•***PCB Layout Design*** – Section *SDIO*: Updated descriptions about the SDIO layout guidelines – Section *Crystal*: Updated descriptions about the crystal layout guidelines
2025-04-02	v1.5	•***Schematic Checklist*** – Section *Strapping Pins*: Updated Table *Boot Mode Control*
2025-01-07	v1.4	•***ESP32-S3 Modules***: – Added download links to module reference designs
2024-11-15	v1.3	•***Schematic Checklist*** – Section *SPI*: Newly added section
2024-01-09	v1.2	•***Schematic Checklist*** – Section *RF Tuning*: Updated RF matching description
2023-12-25	v1.1	•***PCB Layout Design*** – Section *Crystal*: Updated crystal PCB layout
2023-12-22	v1.0	Migrated ESP32-S3 Hardware Design Guidelines from PDF to HTML format. During the migration from PDF to HTML format, minor updates, improvements, and clarifications were made throughout the documentation. If you would like to check previous versions of the document, please submit doc-

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